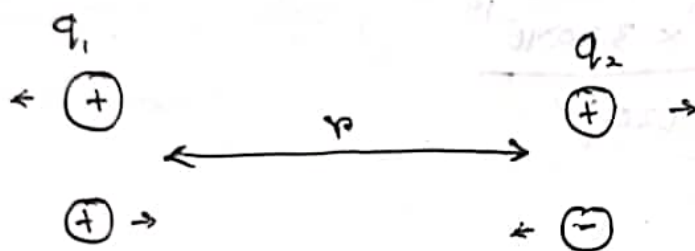


20.03.2023 / Phys

Q

e^-	p^+	ν
$-1.6 \times 10^{-19} \text{ C}$	$+1.6 \times 10^{-19} \text{ C}$	0
$9.11 \times 10^{-31} \text{ kg}$	$1.67 \times 10^{-27} \text{ kg}$	$1.675 \times 10^{-27} \text{ kg}$

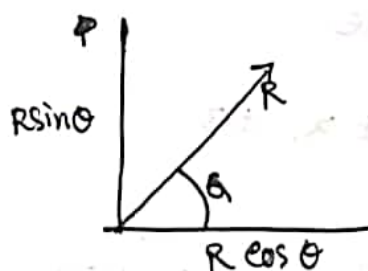
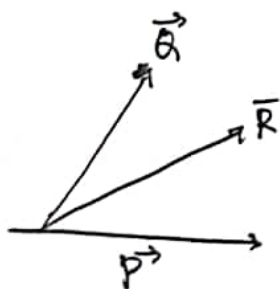
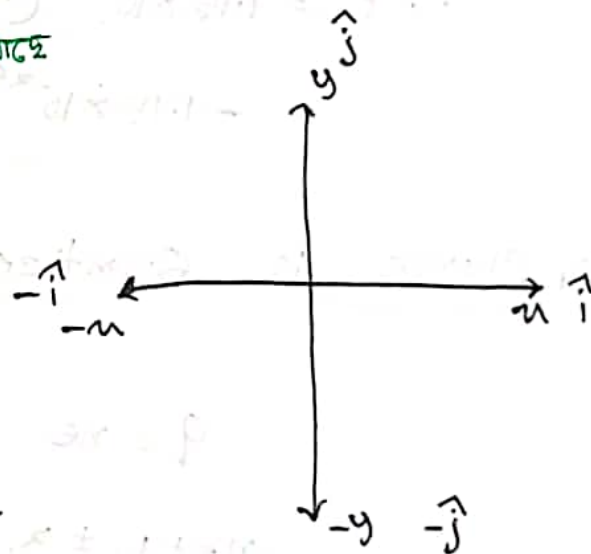
Q



$$\vec{F} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} \hat{r} \rightarrow \text{মান নেই Direction আছে}$$

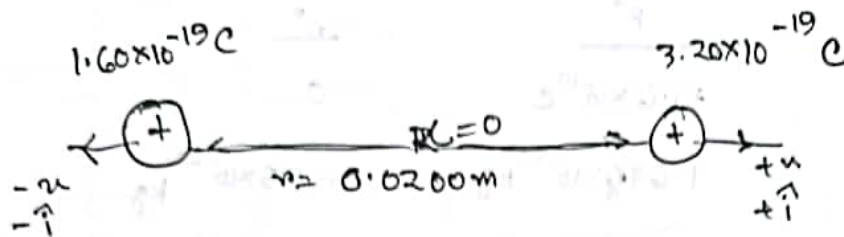
$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$\epsilon_0 = 8.854 \times 10^{-12}$



$$P = 2\hat{i} + 3\hat{j}$$

Q (a) Shows two positively



$$F = 9 \times 10^9 \frac{1.60 \times 10^{-19} \times 3.20 \times 10^{-19}}{(0.0200)^2}$$

$$= 1.15 \times 10^{-24} \text{ N}$$

$$\therefore \vec{F} = 1.15 \times 10^{-24} (-\hat{i})$$

$$= -1.15 \times 10^{-24} \hat{i}$$

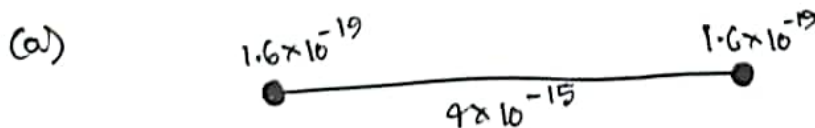
Q Charge is Quantized. $\rightarrow$ e charge $\frac{q}{e}$ must be integer.

$$q = ne$$

$$n = \pm 1, \pm 2, \pm 3$$

$$\frac{6.7 \times 10^{-19} \text{ C}}{1.6 \times 10^{-19} \text{ C}} \rightarrow \text{Quantized } n$$

Q ~~the~~ The nucleus in an iron



$$F = 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{(4 \times 10^{-15})^2} = 14.4 \text{ N}$$

$$= \cancel{14.4 \text{ N}}$$

Direction $\rightarrow$ 180° Angle & opposite direction of each other.

$$(b) F = G \frac{m_1 m_2}{r^2}$$

$$= 6.67 \times 10^{-11} \times \frac{(1.673 \times 10^{-27})^2}{(4 \times 10^{-15})^2}$$

$$= 1.17 \times 10^{-35}$$

2. For Protons the ratio of Electrostatic force (attractive) and Gravitational force < Electrostatic force.

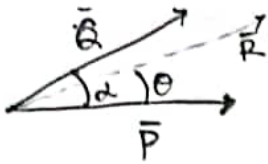
⇒ Charge is Conserved:

$$\gamma \rightarrow e^- + e^+ \quad (\text{Pair production})$$

$$e^- + e^+ \rightarrow \gamma + \gamma \quad (\text{annihilation})$$

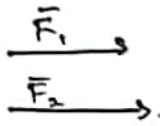
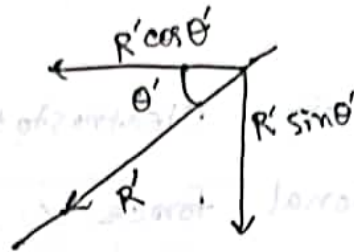
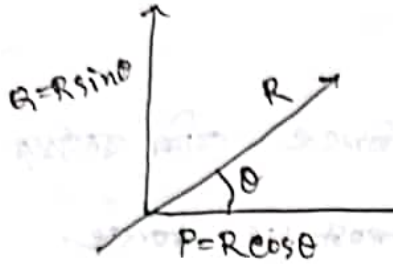
21.03.2023 / Phy

Q

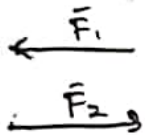


$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

$$\theta = \tan^{-1} \frac{Q \sin \alpha}{P + Q \cos \alpha}$$



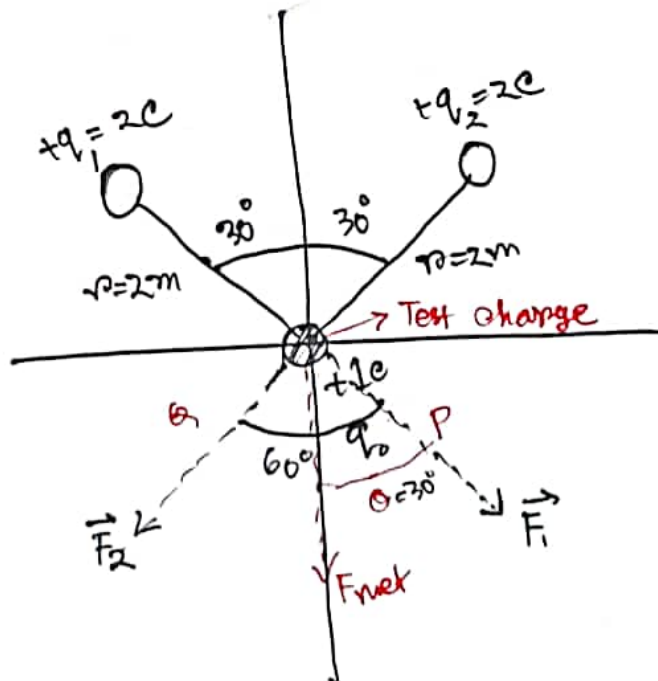
$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2$$



$$\vec{F}_{\text{net}} = \vec{F}_1 - \vec{F}_2$$



Q



+1e ৰাও Force গুণা দিও. অনুভৱ কৰাও।

⑤ Vectors law parallelogram:

$$\Rightarrow F_1 = k \frac{q_1 q_2}{r^2} = 4.5 \times 10^9$$

$$F_2 = k \frac{q_1 q_2}{r^2} = 4.5 \times 10^9$$

$$|\vec{F}_{\text{net}}| = \sqrt{F_1^2 + F_2^2 + 2F_1 F_2 \cos 60^\circ}$$

$$= \sqrt{(4.5 \times 10^9)^2 + (4.5 \times 10^9)^2 + 2 \cdot (4.5 \times 10^9) (4.5 \times 10^9) \cdot \frac{1}{2}}$$

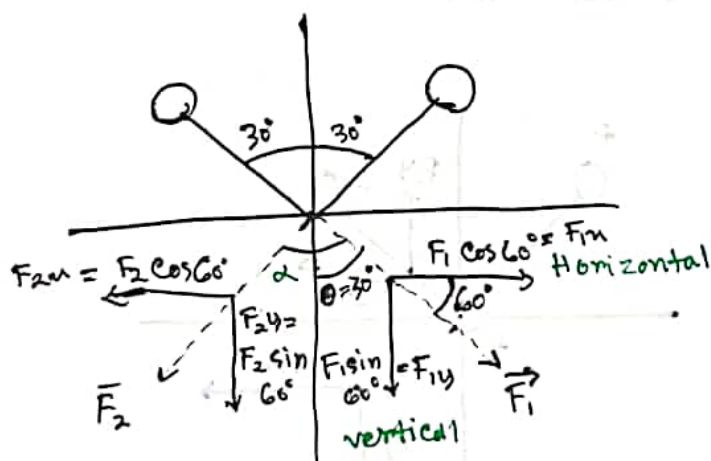
$$= \sqrt{3} (4.5 \times 10^9)$$

$$F_{\text{net}} = 7.79 \times 10^9$$

$$\theta = \tan^{-1} \frac{(4.5 \times 10^9) \sin 60^\circ}{(4.5 \times 10^9) + (4.5 \times 10^9) \cos 60^\circ}$$

$$= 30^\circ$$

⑥ Component split:



$$F_1 = 4.5 \times 10^9$$

$$F_2 = 4.5 \times 10^9$$

$$F_{\text{net horizontal}} = F_1 \cos 60^\circ - F_2 \cos 60^\circ$$

$$= 0 \quad [\text{value same but direction opposite}]$$

$$F_{\text{net vertical}} = F_1 \sin 60^\circ + F_2 \sin 60^\circ$$

$$= 2 F_1 \sin 60^\circ$$

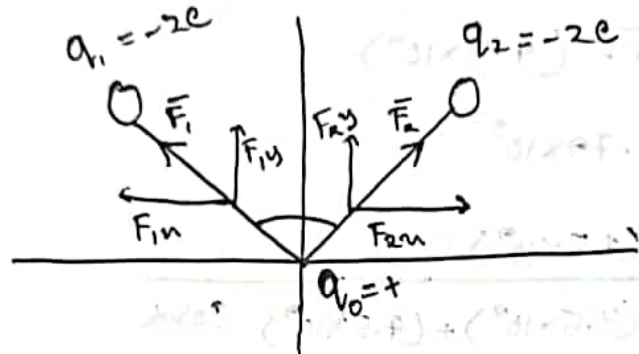
$$= 7.79 \times 10^9$$

$$F_{\text{Net}} = \sqrt{(F_{x\text{Net}})^2 + (F_{y\text{Net}})^2}$$

$$= 7.79 \times 10^9$$

$$\theta = \tan^{-1} \frac{F_{y\text{Net}}}{F_{x\text{Net}}}$$

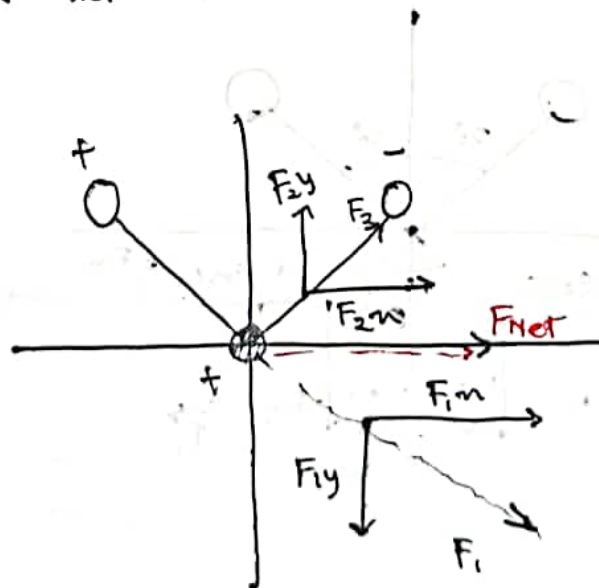
Final net force direction along with negative y-axis.

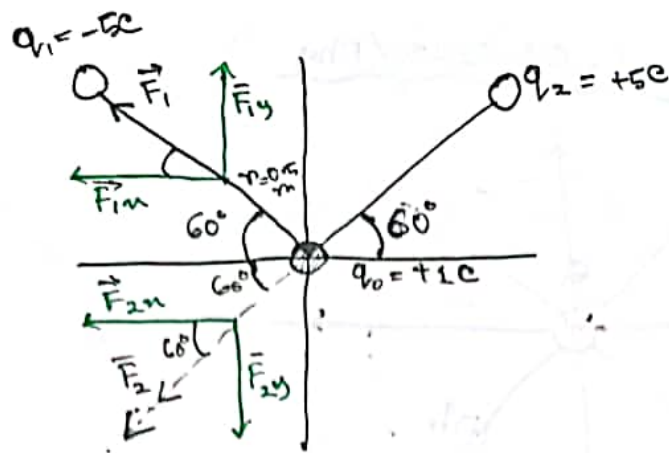


$$F_{x\text{Net}} = F_{1x} - F_{2x}$$

$$F_{y\text{Net}} = F_{1y} + F_{2y}$$

$$F_{\text{Net}} = \sqrt{F_{x\text{Net}}^2 + F_{y\text{Net}}^2}$$





$$F_1 = 9 \times 10^9 \frac{5 \times 1}{0.5} = 9 \times 10^{10}$$

$$F_2 = 9 \times 10^9$$

$$\begin{aligned} F_{\text{Net}} &= F_1 \cos \theta + F_2 \cos \theta \\ &= 2 F_1 \cos 60^\circ \end{aligned}$$

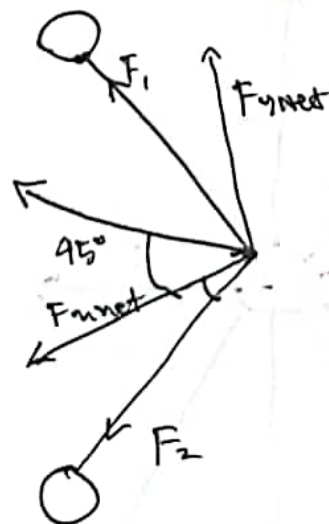
$$\begin{aligned} F_{1y} &= F_1 \sin 60^\circ \\ F_{2y} &= F_2 \sin 60^\circ \end{aligned}$$

$$F_{1y} = F_{2y}$$

$$F_{y\text{Net}} = F_{1y} - F_{2y} = 0$$

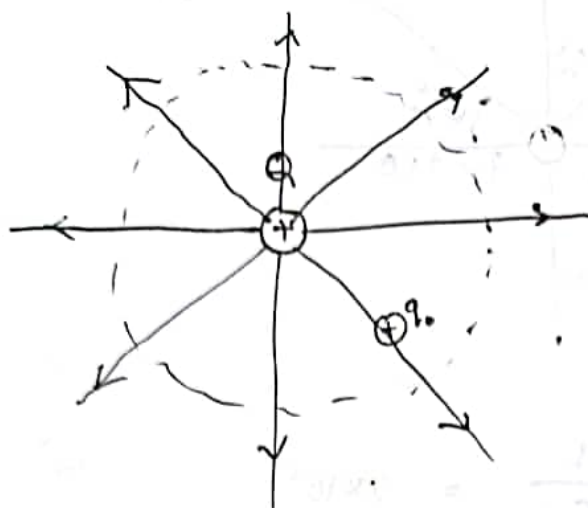
$$F_{\text{Net}} = \sqrt{F_{\text{Net}x}^2 + F_{\text{Net}y}^2}$$

$$\theta = \tan^{-1} \frac{F_{y\text{Net}}}{F_{\text{Net}x}}$$

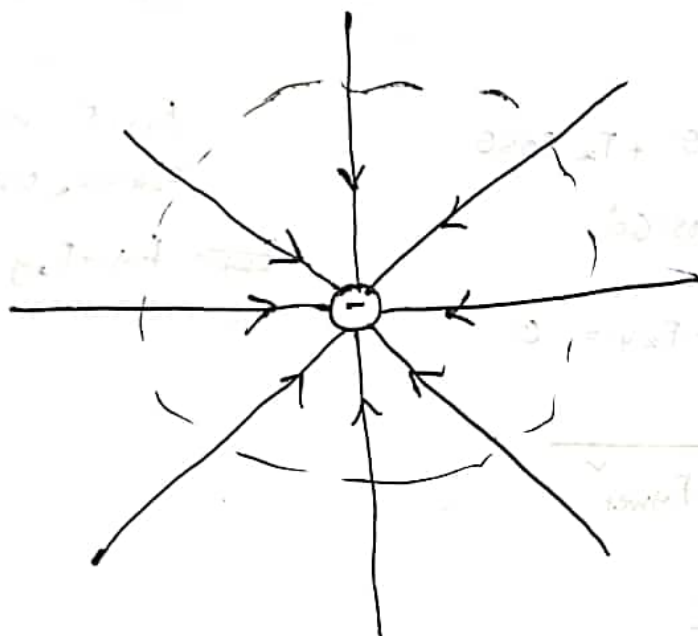


27. 03. 2023 / Phys

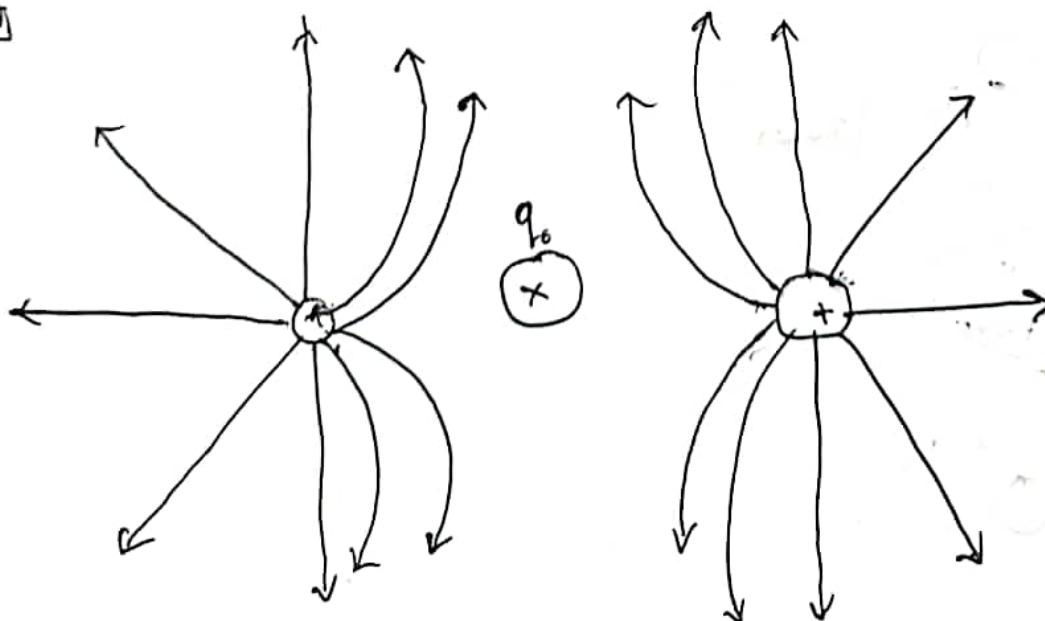
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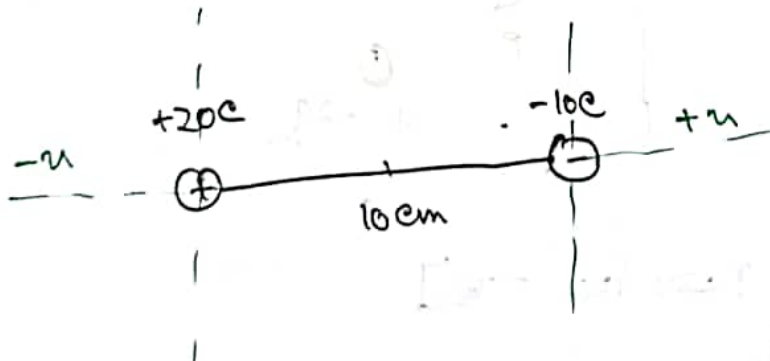
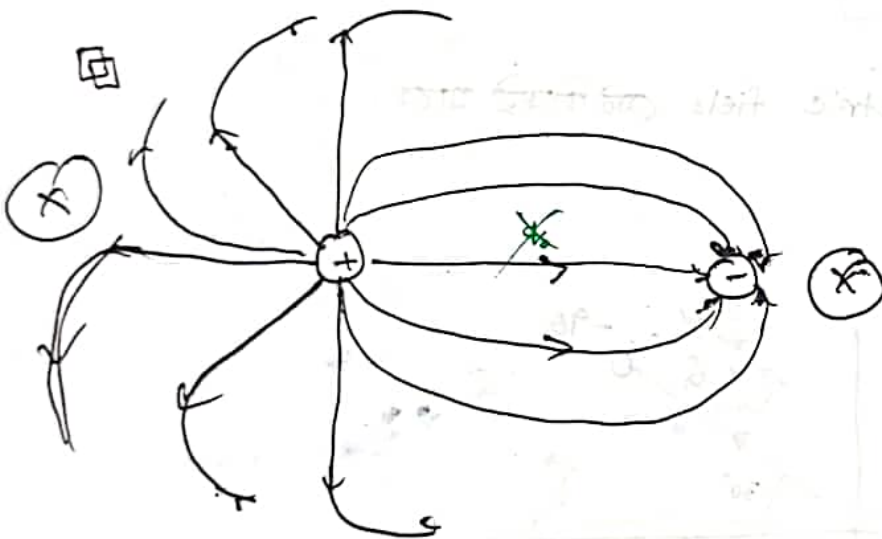
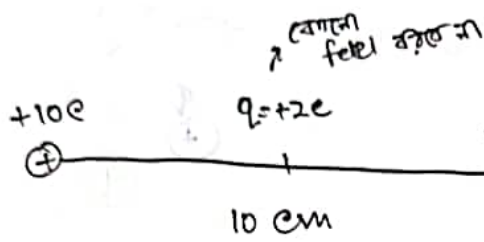
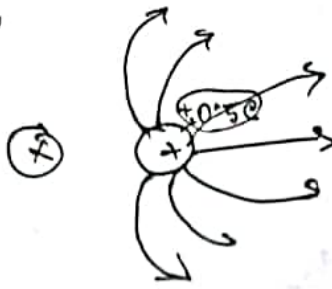
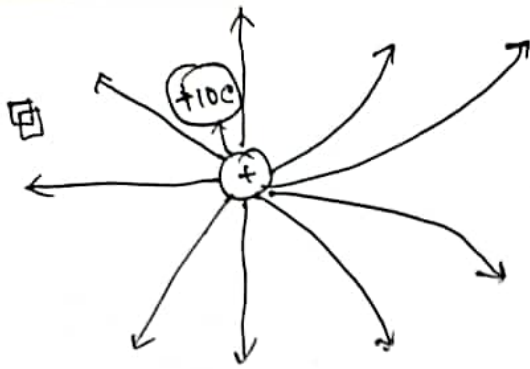


□



□





$$\vec{E} = \frac{\vec{F}}{q_0}$$

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{qq_0}{r^2}$$

$$\therefore E = \frac{F}{q_0}$$

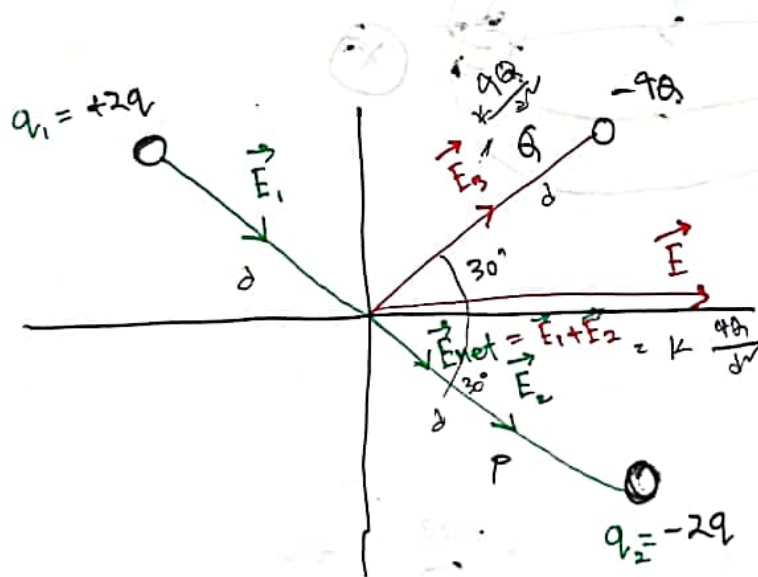
→ Test charge

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cdot q_0}{r^2 q_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

* Electric field generate করে।

□ যেদিকে Force করে Electric field সেই দিকেই থাকে।



$$\vec{E}_{net} = \vec{E}_1 + \vec{E}_2 \quad [এখানে দিচ্ছি পাচ্ছি]$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1}{r_1^2} + \frac{1}{4\pi\epsilon_0} \cdot \frac{q_2}{r_2^2}$$

$$E = \sqrt{p^2 + q^2 + 2pq \cos \alpha}$$

$$\Rightarrow E_1 = k \frac{q_1}{d^2}$$

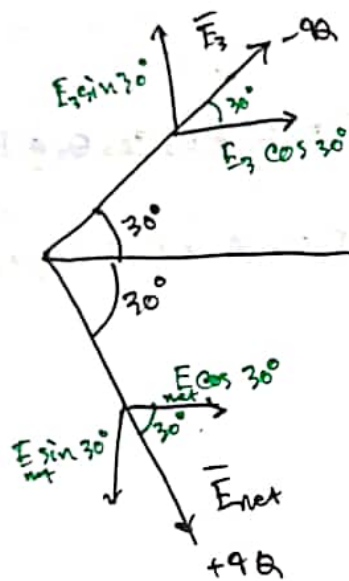
$$E_2 = k \frac{q_2}{d^2}$$

as $\vec{E}_1$ and $\vec{E}_2$ acting at O .

$$E_{\text{net}} = E_1 + E_2 = k \frac{q_1}{d^2} + k \frac{q_2}{d^2}$$

$$= k \frac{2q}{d^2} + k \frac{2q}{d^2}$$

$$= k \frac{4q}{d^2}$$



$$E_3 = k \frac{4q}{d^2}$$

$$E_{\text{net}} = k \frac{4q}{d^2}$$

$$E_{\text{net}} \sin \theta = 0$$

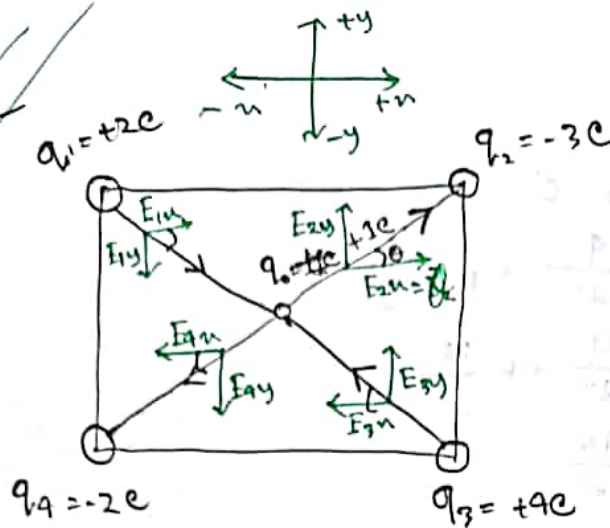
$$\therefore E_{\text{net}} = E_1 \cos 30^\circ + E_2 \cos 30^\circ$$

$$= 2E_1 \cos 30^\circ$$

$$= 2 \times k \frac{4q}{d^2} \cos 30^\circ$$

28.03.2023 / Phys

Formula 1:



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⊙ ←

$$E_n = E_{1n} + E_{2n} - E_{3n} - E_{4n} = E_1 \cos \theta_1 + E_2 \cos \theta_2 - E_3 \cos \theta_3 - E_4 \cos \theta_4$$

$$E_y = -E_{1y} + E_{2y} + E_{3y} - E_{4y} = -E_1 \sin \theta_1 + E_2 \sin \theta_2 + E_3 \sin \theta_3 - E_4 \sin \theta_4$$

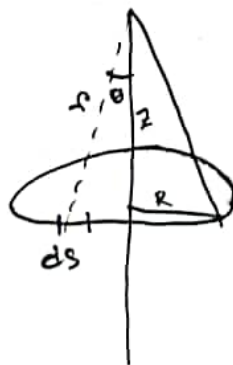
$$E = \sqrt{E_n^2 + E_y^2}$$

Formula 2:

$$E_{13} = E_1 \sim E_3$$

$$E_{24} = E_2 \sim E_4$$

The electric Field Due to a line of charge:



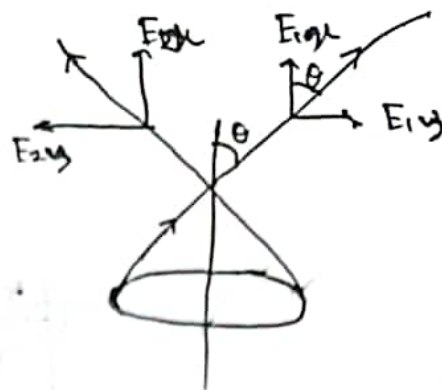
$$\cos \theta = \frac{z}{r}$$

$$r^2 = (z^2 + R^2)$$

$$dq = \lambda ds$$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{(z^2 + R^2)}$$



$$dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{dq}{(z^2 + R^2)} \cdot \frac{z}{r}$$

$$E = \int dE \cos \theta = \int \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)} \cdot \frac{z}{r}$$

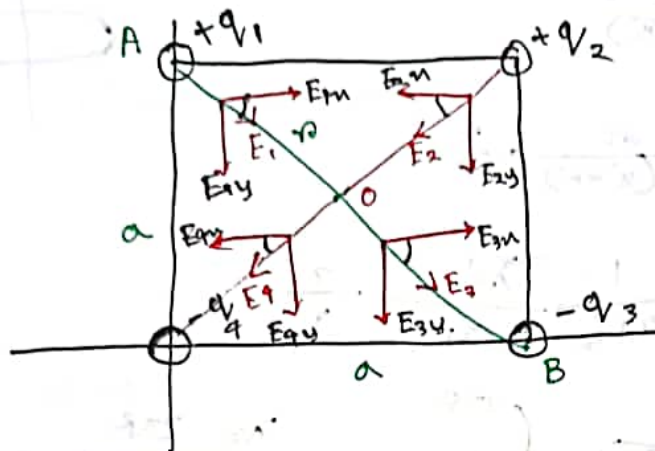
$$= \frac{\lambda}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds = \frac{\lambda z \cdot 2\pi R}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

$$= \int \frac{\lambda ds}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \cdot \frac{z}{r}$$



01.04.2023 / Phys

4



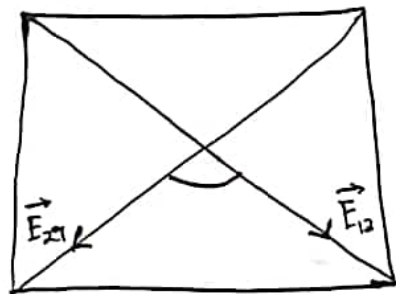
$$\sqrt{2}a = AB$$

$$\frac{AB}{2} = r = \frac{a}{\sqrt{2}}$$

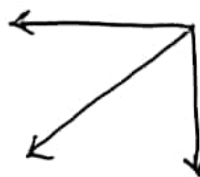
$$\vec{E}_3 = \vec{E}_1 + \vec{E}_2$$

$$\vec{E}_{\text{net}} = \vec{E}_2 + \vec{E}_3$$

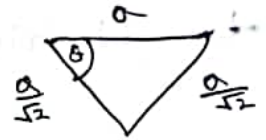
00,



$$R = \sqrt{P^2 + A^2 + 2PA \cos \alpha}$$



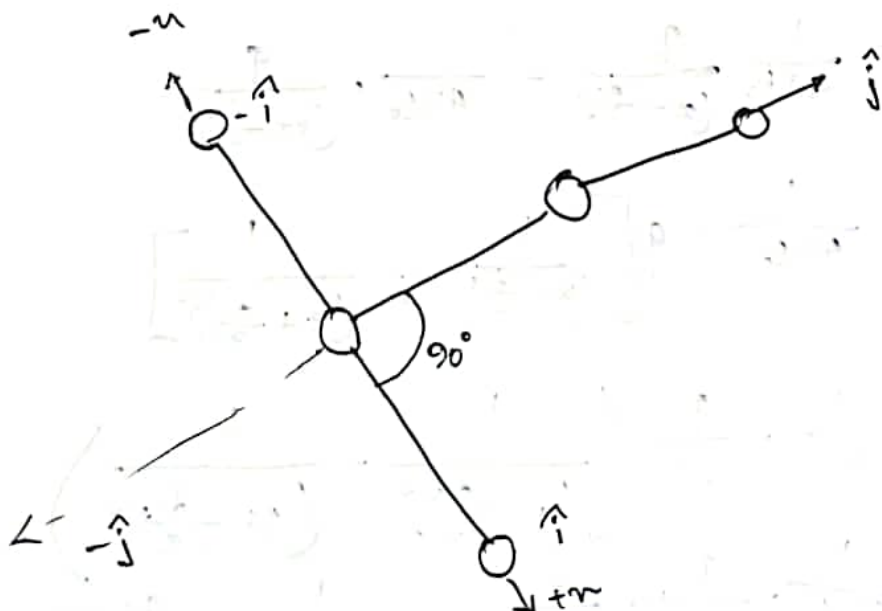
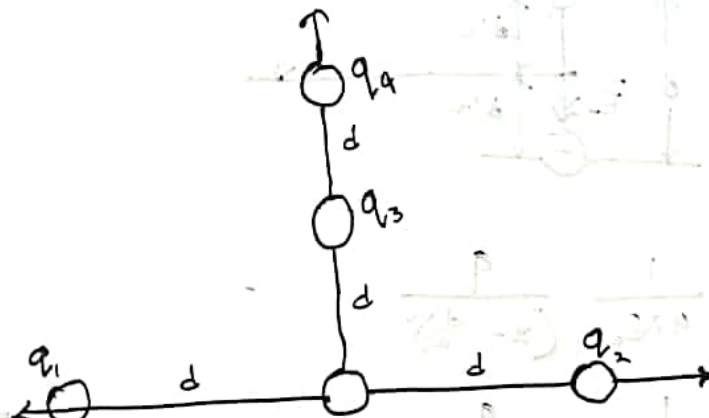
□ ৩ কোণ করত ত্রি



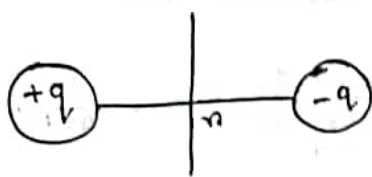
$$\begin{aligned}
 E_{xnet} &= E_{1x} - E_{2x} + E_{3x} - E_{4x} \\
 &= E_1 \cos \theta - E_2 \cos \theta + E_3 \cos \theta - E_4 \cos \theta \\
 &= \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} \cos \theta - \frac{1}{4\pi\epsilon_0} \frac{q_2}{r^2} \cos \theta + \frac{1}{4\pi\epsilon_0} \frac{q_3}{r^2} \cos \theta - \frac{1}{4\pi\epsilon_0} \frac{q_4}{r^2} \cos \theta
 \end{aligned}$$

$$\begin{aligned}
 E_{ynet} &= - (E_{1y} + E_{2y} + E_{3y} + E_{4y}) \\
 &= - \frac{1}{4\pi\epsilon_0 r^2} (q_1 \sin \theta + q_2 \sin \theta + q_3 \sin \theta + q_4 \sin \theta)
 \end{aligned}$$

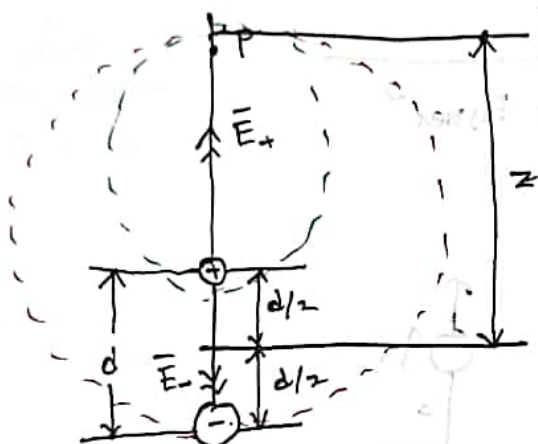
$$\therefore E_{net} = \sqrt{E_{xnet}^2 + E_{ynet}^2}$$



Electric Dipole:



#



$$E_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z - \frac{d}{2}\right)^2}$$

$$E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z + \frac{d}{2}\right)^2}$$

$$E_{\text{net}} = E_+ - E_-$$

$$\therefore E_{\text{net}} = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z - \frac{d}{2}\right)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z + \frac{d}{2}\right)^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot q \left[\frac{1}{\left(z - \frac{d}{2}\right)^2} - \frac{1}{\left(z + \frac{d}{2}\right)^2} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{z^2} \left(\frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{z^2} \cdot \frac{2d}{z} = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3} \rightarrow \text{dipole moment } p$$

$$\left(z - \frac{d}{2}\right)^2$$

$$= \left\{ z \left(1 - \frac{d}{2z}\right) \right\}^2$$

$$= z^2 \left(1 - \frac{d}{2z}\right)^2$$

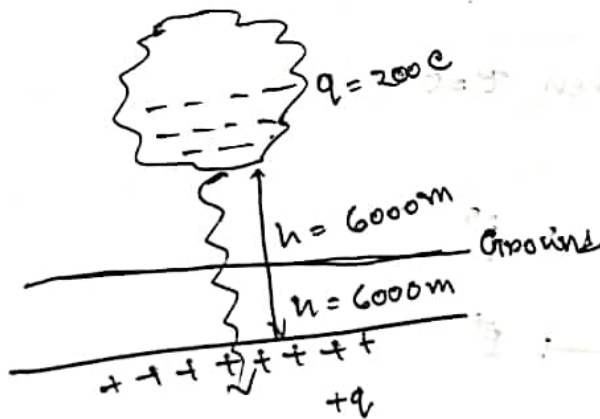
$$\frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \quad 2z \gg d \quad \frac{d}{2z} \ll 1$$

$$E_{\text{net}} = \frac{1}{2\pi \epsilon_0} \cdot \frac{P}{z^3}$$

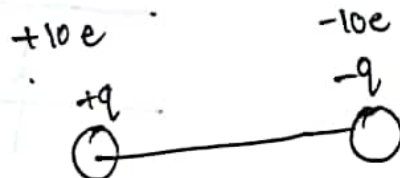
$$E_{\text{net}} \propto P$$



Current flows from the cloud to the ground.



H₂O. 54 (550)

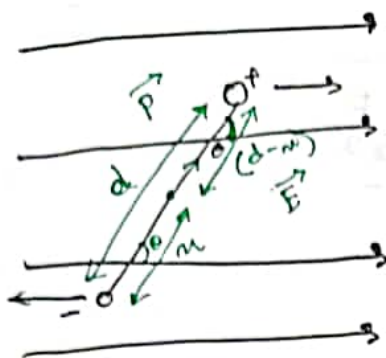


$$P = q \cdot d$$

$$d = \frac{P}{10e}$$

04.04.2023/Phy

⊕



$\tau = p E \sin \theta$

$p = \text{distance}$

$$\tau = F n \sin \theta + F (d - n) \sin \theta$$

$$= F d \sin \theta$$

$$= q E d \sin \theta$$

$$\tau = p E \sin \theta$$

$p = \text{dipole moment}$

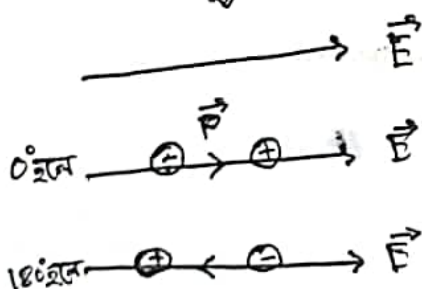
$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$= q d$$

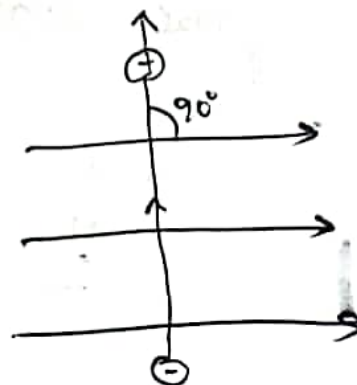
if $\theta = 0^\circ$ or 180° Then $\tau = 0$



#



90° at max torque.



$$\tau = PE \sin \theta \text{ [Anti clockwise]}$$

$$\tau = -PE \sin \theta \text{ [clockwise]}$$

Potential Energy of an Electric Dipole:

$$U = -W = - \int_{90^\circ}^{\theta} \tau d\theta$$

$$= - \int PE \sin \theta d\theta$$

$$\therefore U = -PE \cos \theta$$

$$U = -\vec{P} \cdot \vec{E}$$

For linear

$$W = F \cdot s$$

Electric Potential:



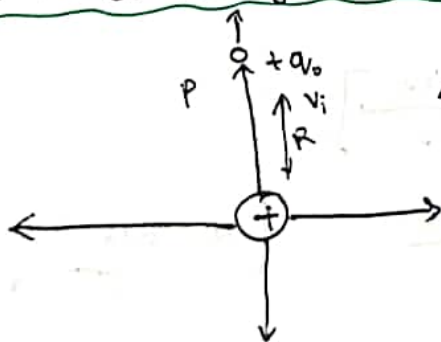
$$W = U = q\Delta V$$

$$\Delta V = V_B - V_A = V_f - V_i$$

$$U = q\Delta V$$

Higher potential क्षेत्र बल दिष्ट Low potential এ তবে বয়ে।

Potential Due to a charged Particle:



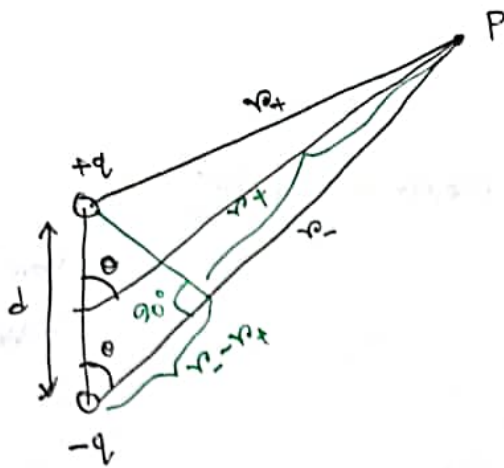
$$\Delta V = W = \int \vec{E} \cdot d\vec{s} = \int E \cos \theta ds$$

$$= \int_R^{\infty} E dr$$

$$= \int_R^{\infty} \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \cdot dr$$

$$= \frac{q}{4\pi\epsilon_0} \int_R^{\infty} \frac{dr}{r^2} \quad \left| \frac{1}{r^2} dr = -\frac{1}{r} \right|$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R}$$



$$V_+ = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r_+}$$

$$V_- = \frac{1}{4\pi\epsilon_0} \cdot \left(\frac{-q}{r_-} \right)$$

$$\therefore V = V_+ + V_-$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_+} - \frac{1}{r_-} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{r_- - r_+}{r_+ r_-} \right]$$

$$\cos\theta = \frac{r_- - r_+}{d}$$

$$\therefore r_- - r_+ = d \cos\theta$$

$$\therefore V = \frac{q}{4\pi\epsilon_0} \left[\frac{r_- - r_+}{r_+ r_-} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \cdot \frac{d \cos\theta}{r^2}$$

[r_- or r_+ or difference $2r$]
 $\frac{1}{\sqrt{2}} \cdot r \cdot r = r^2$

$$\therefore V = \frac{p \cos\theta}{4\pi\epsilon_0 r^2}$$

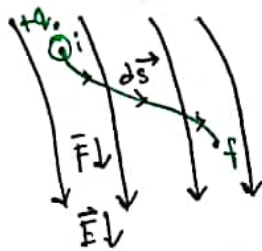
08.04.2023 / PWS

$$\boxed{U = -W}$$
$$\boxed{V = \frac{-W}{q_0} = \frac{U}{q_0}}$$

$$\boxed{\therefore U = q_0 V} \quad [q_0 = +1 \text{ என } U = V]$$

$$\boxed{E = \frac{F}{q_0} = +1e \Rightarrow E = F}$$

Calculating the potential from the Field:



$$V = \frac{-W}{q_0}$$

Now, $dW = \vec{F} \cdot d\vec{s}$

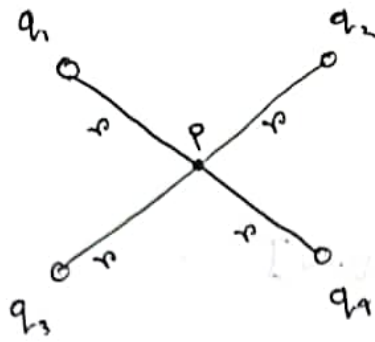
$$= q_0 \vec{E} \cdot d\vec{s}$$

$$W = q_0 \int_i^f \vec{E} \cdot d\vec{s}$$

$$\Rightarrow W = q_0 \int_i^f \vec{E} \cdot d\vec{s}$$

$$\Rightarrow V_f - V_i = \frac{-W}{q_0} = \frac{-q_0 \int_i^f \vec{E} \cdot d\vec{s}}{q_0}$$

$$\boxed{\therefore V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}}$$



$$V_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1}{r}$$

$$V_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_2}{r}$$

$$V_3 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_3}{r}$$

$$V_4 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_4}{r}$$

$$\therefore V = V_1 + V_2 + V_3 + V_4$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon_0 r} [q_1 + q_2 + q_3 + q_4] \quad \begin{array}{l} (+ \text{ sign } \text{शामिल } (+) \\ (- \text{ n } \text{शामिल } (-) \text{ sign } \text{काट } \text{में}) \end{array}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0 r} \cdot [12 - 29 + 31 + 17] \times 10^{-9} \text{ C}$$

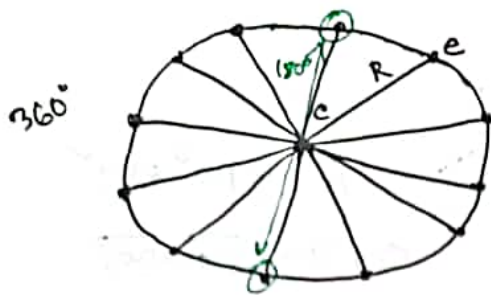
$$= \frac{\sqrt{2}}{4\pi\epsilon_0 \times 1.3} \times 36 \times 10^{-9}$$

$$\therefore V \approx 350 \text{ V}$$

$$r = \frac{d}{\sqrt{2}}$$

$$= \frac{1.3}{\sqrt{2}}$$

$$= 0.919$$



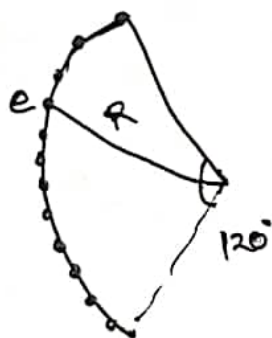
Potential at the centre.

$$V = -12 \cdot \frac{1}{4\pi\epsilon_0} \cdot \frac{e}{R}$$

[electron এর জন্য (-) sign]

এখানে $\vec{E} = 0$ Because প্রতিটি e এর opposite এ 180° Angle

এ আর একটি electron আছে। তাই তারা একে অপরের Canceled out করে।



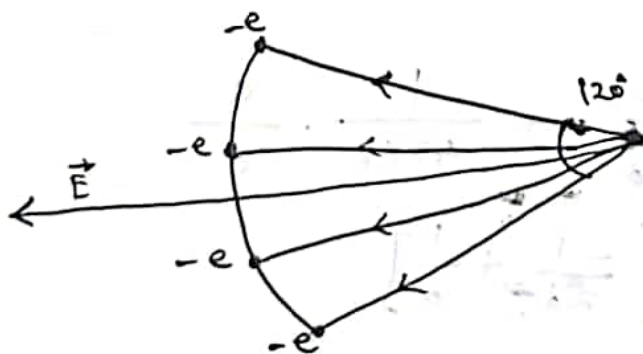
$$V + V + V + V + V = V = V$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{e}{R}$$

V = same

$$\left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c} - \frac{1}{d} \right] \cdot \frac{P}{2\pi R} = V$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{e}{R}$$

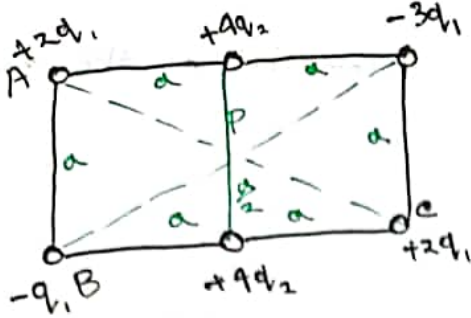


$$\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$$

$$\frac{P}{2\pi R} = V$$

$$\left[\frac{1}{a} + \frac{1}{b} \right] \cdot \frac{P}{2\pi R} = V$$

Example 16:



$$+4q - 4q = 0$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{4q_2}{a/2} + \frac{1}{4\pi\epsilon_0} \cdot \frac{4q_2}{a/2} = \frac{16q_2}{4\pi\epsilon_0 a} = 2.21 \text{ V}$$

Or,

$$V = V_1 + V_2 + V_3 + V_4 + V_5 + V_6 \\ = 2.21 \text{ V}$$

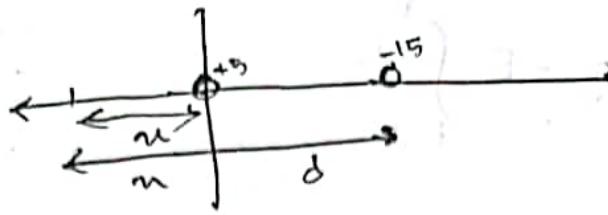
Example 17:

$$V = \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{2d} - \frac{1}{d} + \frac{1}{d} + \frac{1}{d} \right] \\ = 5.62 \times 10^{-4} \text{ V}$$

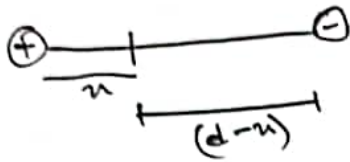
Or,

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{-q}{2d} + \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{d} \\ = \frac{q}{4\pi\epsilon_0} \left[\frac{-1}{2d} + \frac{1}{d} \right] \\ = \frac{q}{4\pi\epsilon_0} \left[\frac{-1+2}{2d} \right] = \frac{q}{8\pi\epsilon_0 d}$$

Example 19:



(a)



$$V(n) = K \left(\frac{q_1}{d_1} + \frac{q_2}{d_2} \right) = 0$$

$$\Rightarrow \frac{q_1}{n} + \frac{q_2}{(d-n)} = 0$$

$$\Rightarrow \frac{5e}{n} - \frac{15e}{(d-n)} = 0$$

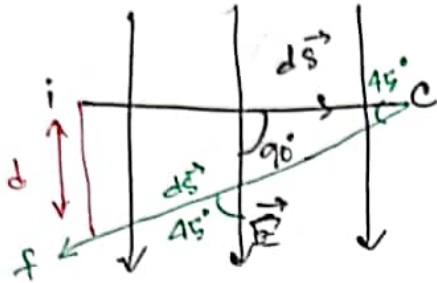
$$\Rightarrow \frac{1}{n} - \frac{3}{(d-n)} = 0$$

$$n = \frac{d}{4} = \frac{24.0}{4} = 6.00 \text{ cm}$$

10 / 09. 2023 / Phy

$$\boxed{\Delta V = V_f - V_i = -Ed = -E \int_i^f ds}$$

$\boxed{\Delta V}$



$$\begin{aligned}\vec{E} \cdot d\vec{s} \\ &= E \, ds \cos 90^\circ \\ &= 0\end{aligned}$$

$$\begin{aligned}\vec{E} \cdot d\vec{s} \\ &= E \, ds \cos 45^\circ\end{aligned}$$

$$\begin{aligned}\Delta V &= - \int_c^f \vec{E} \cdot d\vec{s} \\ &= - \int_c^f E \, ds \cos 45^\circ \\ &= -E \cos 45^\circ \int_c^f ds\end{aligned}$$

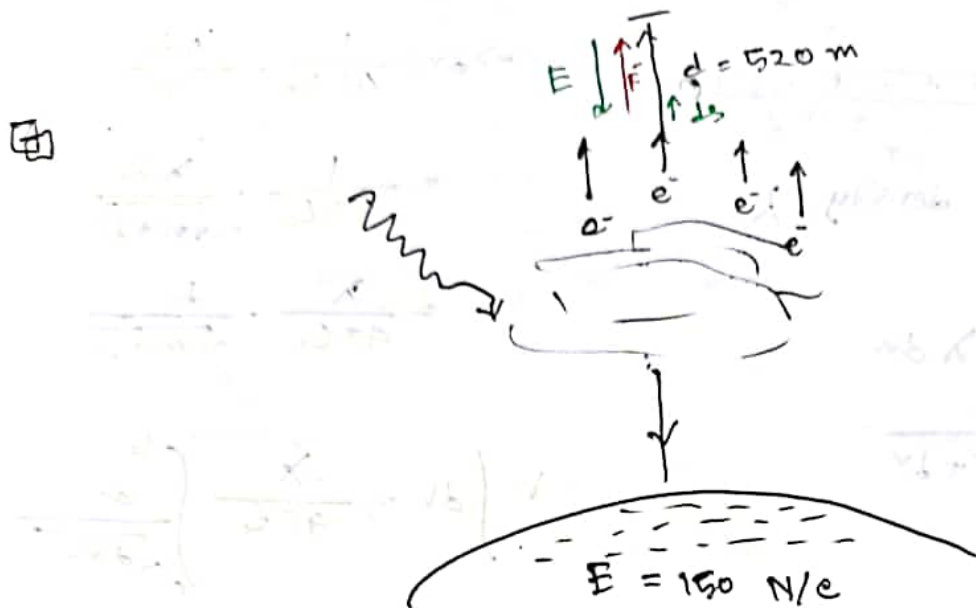
$$\cos 45^\circ = \frac{d}{cf}$$

$$\Rightarrow cf = \frac{d \cancel{\cos 45^\circ}}{\cos 45^\circ}$$

$$\Rightarrow \int_c^f ds = \frac{d}{\cos 45^\circ}$$

$$\text{Now, } V_f - V_i = -E \cos 45^\circ \frac{d}{\cos 45^\circ}$$

$$\therefore V_f - V_i = -Ed$$



$$W = \vec{F} \cdot d\vec{s}$$

$$dW = q\vec{E} \cdot d\vec{s}$$

$$\therefore W = (-1.6 \times 10^{-19}) E \int_0^d ds \cos 180^\circ$$

$$= (-1.6 \times 10^{-19}) (150) (520) \cos 180^\circ$$

$$= 1.2 \times 10^{-19} \text{ J}$$

$$\Delta U = -W$$

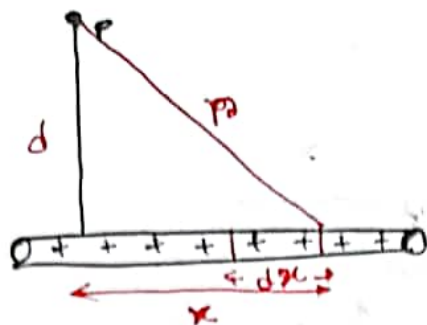
$$= -1.2 \times 10^{-19} \text{ J}$$

$$\Delta V = \frac{-W}{q} = \frac{-1.2 \times 10^{-19}}{-1.6 \times 10^{-19}}$$

$$= 0.75 \text{ V}$$

$$= 75 \text{ kV}$$

Q P



linear charge density λ

$$dq = \lambda du$$

$$r = \sqrt{u^2 + d^2}$$

$$dV = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r}$$

$$\rightarrow dV = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda du}{r}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda du}{\sqrt{u^2 + d^2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{du}{\sqrt{u^2 + d^2}}$$

$$V = \int dV = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{du}{\sqrt{u^2 + d^2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln(u + \sqrt{u^2 + d^2}) \right]_0^L$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln(L + \sqrt{L^2 + d^2}) - \ln d \right]$$

$$\therefore V = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{L + \sqrt{L^2 + d^2}}{d}\right)$$

Assignment:

$$\vec{F}_1 = 3\hat{i} - 4\hat{j}$$

$$|\vec{F}_1| = 5$$

$$\vec{F}_2$$

$$\theta = 60^\circ$$

$$40\text{ N}$$

Charged Disk:

$$V = \int dV = \frac{\sigma}{2\epsilon_0} \int_0^R \frac{R' dR'}{\sqrt{z^2 + (R')^2}}$$

$$= \frac{\sigma}{2\epsilon_0} \int_{z^2}^{z^2 + R^2} \frac{du}{2\sqrt{u}}$$

$$= \frac{\sigma}{4\epsilon_0} \int_{z^2}^{z^2 + R^2} x^{-\frac{1}{2}} du$$

$$= \frac{\sigma}{4\epsilon_0} \left[\frac{x^{-\frac{1}{2} + 1}}{-\frac{1}{2} + 1} \right]_{z^2}^{z^2 + R^2}$$

$$= \frac{2\sigma}{4\epsilon_0} \sqrt{x} \Big|_{z^2}^{z^2 + R^2}$$

$$= \frac{\sigma}{2\epsilon_0} \left[\sqrt{z^2 + R^2} - z \right]$$

$$z^2 + R'^2 = x$$

$$\Rightarrow 2R' dR' = dx$$

$$\Rightarrow R' dR' = \frac{dx}{2}$$

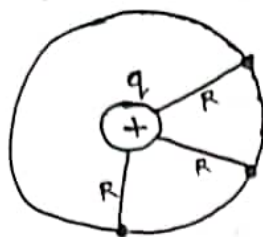
$$\text{When } R' = 0$$

$$x = z^2$$

$$\text{When } R' = R$$

$$x = z^2 + R^2$$

11.04.2023 / PM4



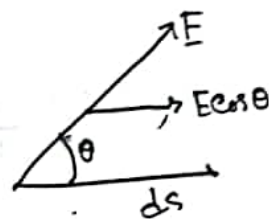
$$dV = \frac{-dW}{q_0}$$

$$\Rightarrow dW = -q_0 dV$$

$$\Rightarrow dW = q_0 (E \cos \theta) ds$$

$$\therefore -q_0 dV = q_0 (E \cos \theta) ds$$

$$\therefore E = \frac{-dV}{ds}$$



$$\square U = q_0 V$$

$$= q_0 \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

$$U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cdot q_0}{r}$$

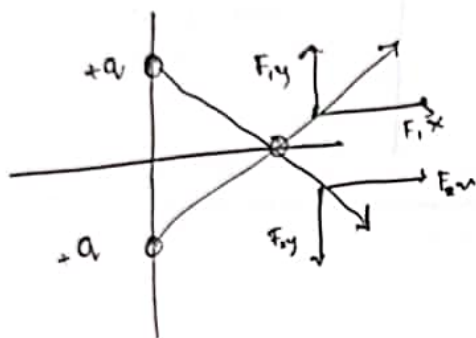
$$\text{any, } F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cdot q_0}{r^2}$$

29.04.2023 / PhM

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{r} (q_1 + q_2 + q_3 + q_4) \quad [\text{when } r \text{ same}]$$

□



$$F_{\text{net}} = F_{1x} + F_{2x}$$

$$F_{\text{net}} = \sqrt{F_{1x}^2 + F_{2x}^2}$$

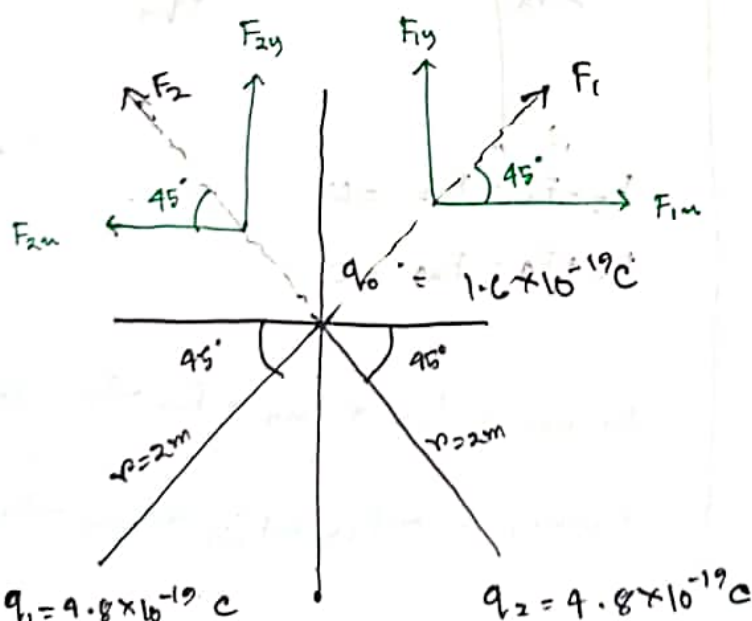
$$F_{\text{net}} = F_{\text{net}}$$

$$\vec{F}_{y\text{net}} = 2.94 \times 10^{-28} \hat{j}$$

$$F_1 = \frac{9 \times 10^9 \times 4.8 \times 10^{-19} \times 1.6 \times 10^{-19}}{2^2} = 1.728 \times 10^{-28} \text{ N}$$

$$F_2 = \frac{9 \times 10^9 \times 1.6 \times 10^{-19} \times 4.8 \times 10^{-19}}{2^2} = 1.728 \times 10^{-28} \text{ N}$$

$$|F_1| = |F_2|$$

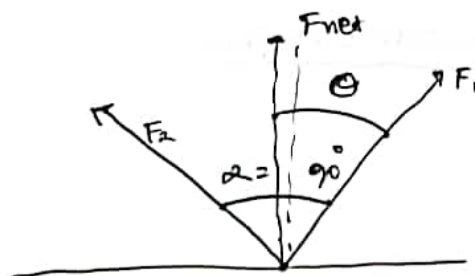


$$F_{\text{net}} = F_{1x} - F_{2x} = F_1 \cos 45^\circ - F_2 \cos 45^\circ = 0$$

$$F_{y\text{net}} = F_{1y} + F_{2y} = F_1 \sin 45^\circ + F_2 \sin 45^\circ = F_1 \sin 45^\circ + F_1 \sin 45^\circ = 2F_1 \sin 45^\circ$$

$$\therefore F_{\text{net}} = F_{y\text{net}} = 2.94 \times 10^{-28} \text{ (positive y axis direction)} = 2 \times 1.728 \times 10^{-28} \times \sin 45^\circ = 2.44 \times 10^{-28}$$

Or

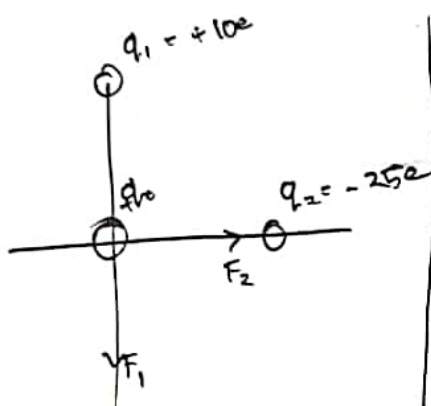


$$F_1 = F_2 = 1.78 \times 10^{-28} \text{ N}$$

$$F_{\text{net}} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 90^\circ}$$

$$\tan \theta = \frac{F_2 \sin 90^\circ}{F_1 + F_2 \cos 90^\circ}$$

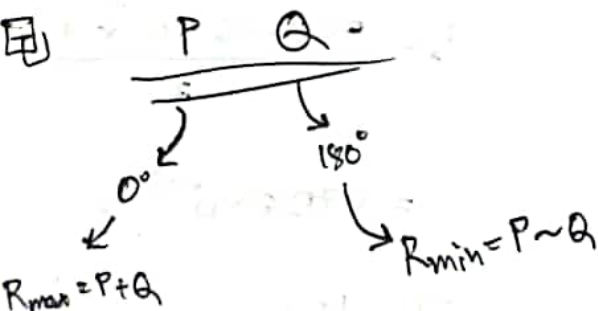
Q



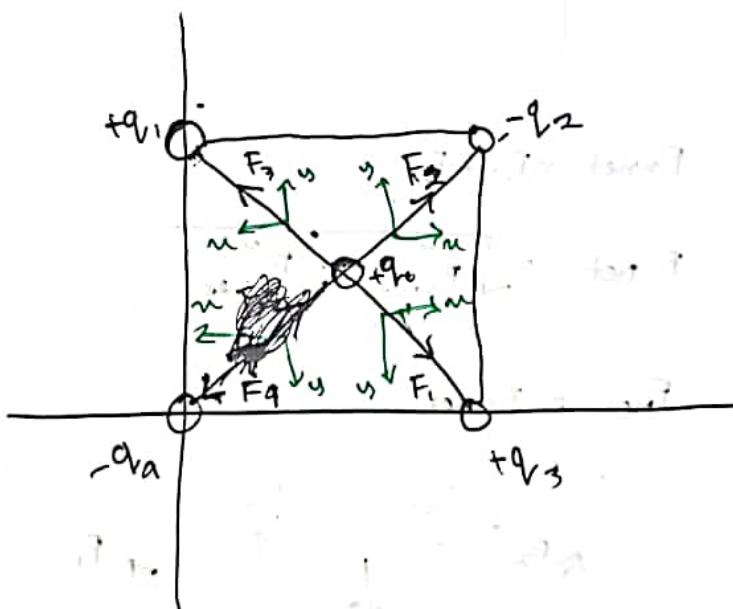
$$F_{\text{net}} = F_{2x} + F_{1x}$$

$$F_{y\text{net}} = F_{2y} + F_{1y}$$

Q



Q



$$F_3 - F_1 = F_{13} = 0$$

$$F_2 + F_4 = F_{24} = 0$$

$$F_{\text{net}} = F_{1x} + F_{2x} - F_{3x} - F_{4x} = 0$$

$$F_{y\text{net}} = -F_{1y} + F_{2y} + F_{3y} - F_{4y} = 0$$